

Samir Si-Mohammed

PhD, Associate Professor

Born on August 2nd, 1998. Algerian.

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Current Situation

- 09/2025 - : **Associate Professor**, *Université de Lorraine*, CNU Section 61, France.
- **Lab** : Centre de Recherche en Automatique de Nancy (CRAN).
 - **Teaching institution** : Polytech Nancy.
 - **Research Interest**: Internet of Things, Wireless Networks, Experimental Research, Digital Twins.

Work Experiences

- 10/2023 - 09/2025: **Postdoctoral Researcher**, *University of Strasbourg*, ICube laboratory, France.
- **Research Interest**: Internet of Things, Wireless Networks, Experimental Research, Digital Twins.
 - **Advisor**: Dr. Fabrice Théoleyre.
- 09/2020 - 09/2023: **PhD Student**, *École Normale Supérieure de Lyon*, LIP laboratory, France.
- **Topic**: Internet of Things, Performance Evaluation, Wireless Networks.
 - **Advisor**: Prof. Thomas Begin, Prof. Isabelle Guérin Lassous, Dr. Pascale Vicat-Blanc.
- 01/2020 - 06/2020 : **Research Internship**, *EURECOM*, Sophia-Antipolis, France.
- **Topic**: Orchestration and Optimization of UAV Flights in 5G Networks.
 - **Advisor**: Prof. Adlen Ksentini.
- 09/2018 : **Internship**, *Ericsson*, Algiers, Algeria.
- **Topic**: Design of a technical and commercial offer for Algérie Télécom on IMS Core Network.
 - **Advisors**: Wladimiro Storti, Yassine Zerrouki.

Education & Degrees

- 2020 - 2023 : **PhD, Computer Science**, *École Normale Supérieure de Lyon*, Lyon, France.
- **Topic**: Multi-criteria Selection and Configuration of IoT Network Technologies.
 - **Advisors**: Thomas Begin, Isabelle Guérin Lassous, Pascale Vicat-Blanc.
 - **Jury**: Fabrice Théoleyre, Thomas Watteyne, Alexandre Guitton, Angela Nicoara.
- 2019 - 2020 : **Master of Research, Computer Science**, *École nationale Supérieure d'Informatique*, Algiers, Algeria.
- **Topic**: Study of Deep Learning techniques for Tracking in Augmented Reality.
 - **Advisor**: Karima Benatchba.
- 2015 - 2020 : **Engineer, Computer Science**, *École nationale Supérieure d'Informatique*, Algiers, Algeria.
- **Topic**: Orchestration and optimization of UAV Drone's flights in 5G Networks.
 - **Advisors**: Yacine Challal, Amar Balla.

Research visits

- 10/2025 : **Visiting Researcher (1 week)**, *Pusan National University*, Wireless AI Lab, South Korea.
- **Topic**: Quantum Computing for Wireless Network Optimization.
 - **Correspondent**: Won-Joo Hwang.
- 07/2025 : **Visiting Researcher (2 weeks)**, *University of Malaga*, NICS Lab, Spain.
- **Topic**: Security in Digital Twins.
 - **Correspondents**: Cristina Alcaraz, Javier Lopez.
- 06/2023 - 09/2023 : **Visiting Scholar (3 months)**, *University of Waterloo*, Department of Electrical and Computer Engineering, Canada.
- **Topic**: Internet of Things, 5G, Localization.
 - **Advisor**: Catherine Rosenberg.

International collaborations

- 05/2025 - **Pusan National University**, Prof. Won-Joo Hwang, Pusan, South Korea.
- **Topic:** Quantum annealing for routing in IoT networks.
- 04/2025 - **University of Malaga**, Dr. Cristina Alcaraz & Prof. Javier Lopez, Malaga, Spain.
- **Topic:** Providing security in Digital Twins.
- 09/2023 - **University at Buffalo**, Dr. Filippo Malandra, New York, USA.
- **Topic:** No-code simulation for IoT networks in ns-3

Publications

International Journals

- 2026 : Samir Si-Mohammed and Fabrice Théoleyre. *Group-Based Link Modeling for Wireless Digital Twins: Towards Accurate Network Performance Prediction*. **Elsevier Internet of Things Journal**.
- 2024 : Samir Si-Mohammed, Anthony Bardou, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *NS+NDT: Smart Integration of Network Simulation in Network Digital Twin, Application to IoT Networks*. **Future Generation Computer Systems**, volume 157, pages 124–144.
- 2024 : Mohammad Abuyaghi, Samir Si-Mohammed, George Shaker, and Catherine Rosenberg. *Positioning in 5G Networks: Emerging Techniques, Use Cases, and Challenges*. **IEEE Internet of Things Journal**.
- 2023 : Samir Si-Mohammed, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *HINTS: A Methodology for IoT Network Technology and Configuration Decision*. **Elsevier IoT Journal**.
- 2021 : Samir Si-Mohammed, Maha Bouaziz, Hamed Hellaoui, Oussama Bekkouche, Adlen Ksentini, Tarik Taleb, Lechoslaw Tomaszewski, Thomas Lutz, Gokul Srinivasan, Tanel Jarvet, and Pawel Montowtt. *Supporting Unmanned Aerial Vehicle Services in 5G Networks: New High-Level Architecture Integrating 5G With U-Space*. **IEEE Vehicular Technology Magazine**, volume 16, pages 57–65.

International Conferences

- 2026 : Ghinwa Ismail, Samir Si-Mohammed and Fabrice Theoleyre. *Lightweight Trace-Driven Burst Traffic Generation for 5G Network Digital Twins*. In **NetSoft 2026 - 2026 IEEE Conference on Network Softwarization (to appear)**.
- 2025 : Samir Si-Mohammed and Fabrice Théoleyre. *Per Link Data-driven Network Replication Towards Self-Adaptive Digital Twins*. In **MSWiM 2025 - 2025 IEEE International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems**.
- 2025 : Samir Si-Mohammed and Fabrice Théoleyre. *Data-Driven Prediction Models for Wireless Network Configuration*. In **AINA 2025 - 2025 International Conference on Advanced Information Networking and Applications**.
- 2024 : Nicholas Accurso, Samir Si-Mohammed, Diptangshu De, and Filippo Malandra. *WTTool: A Visual Web-based Topology Generator and 5G Network Simulator with ns-3 (demo)*. In **CloT 2024 - 2024 Conference on Cloud and Internet of Things 2024**.
- 2023 : Samir Si-Mohammed, Zakaria Fraoui, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *StackNet: IoT Network Simulation as a Service*. In **ICC 2023 - 2023 IEEE International Conference on Communications**.
- 2022 : Samir Si-Mohammed, Malasri Janumporn, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *SIFRAN: Evaluating IoT Networks with a no-code Framework based on ns-3*. In **ACM LANC 2022 - Latin American Networking Conference**.

- 2022 : Samir Si-Mohammed, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *ADlperf: A Framework for Application-driven IoT Network Performance Evaluation*. In **ICCCN 2022 - 2022 IEEE International Conference on Computer Communications and Networks**.
- 2020 : Samir Si-Mohammed, Adlen Ksentini, Maha Bouaziz, Yacine Challal, and Amar Balla. *UAV mission optimization in 5G: On reducing MEC service relocation*. In **GLOBECOM 2020 - 2020 IEEE Global Communications Conference**.

National Conferences

- 2026 : Samir Si-Mohammed and Fabrice Théoleyre. *Réplication de Réseaux par Lien pour des Jumeaux Numériques Adaptatifs*. In **CoRes 2026 (to appear)**.
- 2025 : Samir Si-Mohammed and Fabrice Théoleyre. *Modèles de prédiction basés sur des données pour la configuration des réseaux sans-fil*. In **CoRes 2025**.
- 2023 : Samir Si-Mohammed, Thomas Begin, Isabelle Guérin Lassous, and Pascale Vicat-Blanc. *COSIMIA : Combiner Simulation et Apprentissage Automatique pour l'Optimisation des Configurations Réseau IoT*. In **CoRes 2023**.

White Papers

- 2025 : Mathieu Bacou, David Beserra, Eugen Dedu, Loïc Desgeorges, Didier Donsez, Alexandre Guitton, Baptiste Jonglez, Arnaud Legrand, Georgios Papadopoulos, Olivier Richard, Samir Si-Mohammed, Nina Tamdrari, and Fabrice Theoleyre. Journée thématique du GDR RSD : Pratiques expérimentales de la communauté systèmes & réseaux.

Scientific Animation

Technical Program Committee

- 2026 : - International Conference on Telecommunications (ICT).
- 2026 : - URBan SENSEmaking and Intelligence for Safer Cities (URBSENSE).
- 2025 : - IEEE Vehicular Technology Conference (VTC).
- 2024/2025 : - IEEE Symposium on Computers and Communications (ISCC).
- 2024 : - IEEE International Conference on Network Protocols (ICNP) Posters/Demos session.

International Journals

- ACM Computing Surveys.
- Elsevier Computer Networks.
- Elsevier Computer Communications.
- Elsevier Ad Hoc Networks.
- ITU Journal on Future and Evolving Technologies.
- IEEE Access.
- IEEE Networking Letters.
- IEEE Communications Magazine.
- SAGE Structural Health Monitoring.
- IEEE Network Magazine.

International Conferences

- International Conference on Computing, Networking and Communications (ICNC).
- IEEE International Conference on Communications (ICC).
- IEEE International Conference on Pervasive Computing and Communications - Workshops and Affiliated Events - (HCCS).

Research Supervision

- 11/2024 - * : **Ghinwa Ismail**, *University of Strasbourg*, PhD thesis.
Digital Twins for Efficient Wireless Networks. Co-supervision with Fabrice Théoleyre.
- 01/2026 - * : **Rahma Mihoub**, *CRAN & ESTIN (Bejaia, Algeria)*, Final year Internship.
Calibration of IoT simulation through real experimentation.
- 01/2026 - * : **Mohamed Ouaddane**, *CRAN & ESTIN*, Final year Internship.
Knowledge Distillation for complexity reduction in ML models for networks. Co-supervision with Moufida Maimour.
- 04/2026 - * : **Ikram Kellou & Aya Boussaha**, *University of Ghardaia*, Final year Internship.
Predictive maintenance for industrial systems using DRL. Co-supervision with Mohamed El Amine Fekair.
- 06/2024 - 08/2024 : **Selma D'Alimonte**, *ENSEEIHT (Bac +4)*, Summer Internship.
Build the Digital Twin of a Wi-Fi 6 (IEEE 802.11ax) network. Co-supervision with Fabrice Théoleyre.
- 06/2021 - 09/2021 : **Malasri Janumporn**, *Institut Universitaire de Technologie (IUT) Lyon 1 (Bac+2)*, Internship.
Development of a web platform for the interactive use of ns-3 scripts. Co-supervision with Thomas Begin.

Teaching

At Polytech Nancy, I'm the pedagogical responsible for the courses of **Network Administration — SDN**, **Network Infrastructures** and **Wireless Networks**, for Bac+4 and Bac+5 (last year) engineering students at the IA2R departement. Prior to that, I delivered the following teachings:

- Spring 2024/2025 : **Wireless Networks**, *University of Strasbourg (Bac+4)*.
Course Responsible.
Volume: 22h (Lectures + Exercise/Practical sessions).
- Spring 2024 : **Systems Programming**, *University of Strasbourg (Bac+2)*.
- **Responsible:** Benoît Naegel.
- **Volume:** 20h (Practical sessions).
- Spring 2024 : **Local Networks**, *University of Strasbourg (Bac+3)*.
- **Responsible:** Pascal Mérimol.
- **Volume:** 8h (Lectures + Exercise sessions).
- Fall 2021 : **Performance Evaluation in Networks**, *École Normale Supérieure de Lyon (Bac+4)*.
- **Responsible:** Éric Thierry.
- **Volume:** 28h (Practical/Exercise Sessions).
- Fall 2021 : **Introduction to Networks**, *École Normale Supérieure de Lyon (Bac+3)*.
- **Responsibles:** Thomas Begin, Isabelle Guérin Lassous.
- **Volume:** 32h (Practical/Exercise Sessions).
- Spring 2021 : **Distributed Algorithms**, *University of Lyon 1 Claude Bernard (Bac+4)*.
- **Responsible:** Isabelle Guérin Lassous.
- **Volume:** 27h (Practical/Exercise Sessions).

Invited Talks

- 27/01/2026 : **"IMPACT SO 2026"**, *Invited Speaker*, Inist Nancy, Participation to a "table ronde" around the impact of open science on research professions..

- 2024-25 : **Digital Twins for Self-Configurable Wireless Networks**, Seminar.
- Quantum AI Lab, Pusan National University, South Korea, 10/25
 - NICS lab, University of Malaga, Spain, 07/25
 - HoWNet Team, LIP lab, Lyon, 02/25
 - ARN Team, CRAN lab, Nancy, 01/25
 - FUN Team, Inria Lille, Lille, 12/24
 - NPA Team, LIP6 lab, Sorbonne Université, Paris, 11/24
 - DIANA Team, Inria Sophia-Antipolis (Online), 11/24
- 25/01/2025 : **"Les Entretiens de l'Excellence"**, *Invited Speaker*, EM Strasbourg, Animated a session on Education and Research for high school students..
- 04/2023 : **Towards a Digital Twin coupling Simulation & ML for IoT Networks Design and Optimization**, *Networks Team*, Invited Speaker, ICube Lab, University of Strasbourg.
- 09/2020 : **Orchestration and Optimization of UAV Flights in 5G Networks**, *GT-Eval Perf*, Speaker, CITI Lab, INSA de Lyon.
- 05/2020 : **MSP Unfiltered**, *Microsoft Student Learn Ambassadors program*, Speaker, Online.
- 05/2020 : **All About Microsoft Student Partners**, *Microsoft Student Learn Ambassadors program*, Speaker & Host, Online.
- 04/2020 : **Neural Networks Workshop**, *AI2E School of AI Algiers*, Speaker & Trainer, Online.
- 04/2020 : **Introduction to Python for AI**, *Microsoft Student Learn Ambassadors program*, Speaker, Online.

Software

- Spring 2024 : **WT-Tool, a no-code framework for building and sharing wireless technologies**, *University of Strasbourg, Collaboration with University at Buffalo (USA)*.
- *Used tools*: ns-3, Flask, Docker.
- Summer 2021 : **SIFRAN, a no-code framework for the interactive use of ns-3**, *École Normale Supérieure de Lyon*.
- *Used tools*: ns-3, Flask, Scalingo, MongoDB.
- 09/2019 : **AI model for the Prediction of the outcomes of Italian football league games**, *Disruptive Data Summer School*.
- *Used tools*: Scikit Learn, Python, MySQL.
- Summer 2017 : **JobExpress, a platform for allowing the interaction between workers and individuals**, *Hack!T Hackathon: Special Prize of the Jury*.
- *Used tools*: Laravel, PHP.
- 02/2017 - 06/2017 : **Stud'Up, a desktop application for student agenda**, *École nationale Supérieure d'Informatique*.
- *Used tools*: C#, Visual Studio, MySQL.

Languages

English, French, Arabic: Common.

Spanish: Read, written, spoken.

Amazigh: Native.